

BloodLink project case study

- **What problem did it solve ?**

The biggest problem we solve is the lack of coordination between citizens and hospitals when there's a need for blood donations. 'BloodLink' provides a platform where citizens can be informed at any time by needs for blood donations through urgent alerts or through organised donation campaigns. Our project also offers hospitals the ability to manage and keep track of blood stocks more efficiently as well as organise inter-hospital transfers.

- **Who was it for ?**

Our two main stakeholders are

- **Hospitals** : Hospital staff can consult and manage their hospital's blood stock, create and manage inter-hospital blood stock transfers, manage blood donations and donation campaigns, manage audits & logs and manage donors.
- **Donors** : Create and manage their accounts, check for initial donation eligibility which is later verified by a doctor on-site, donate and participate in donation campaigns.

- **What I personally built ?**

I was personally responsible for developing the blood inventory management feature, which allows hospital staff to consult and manage the blood stock of different blood types respective to their hospital, whilst admins can access the blood stock of all registered hospitals. I also developed the inter-hospital blood transfers feature, enabling hospital staff to create, manage and organise transfer requests between their respective hospitals and other registered ones while the admins have access to all transfer requests between all hospitals. Additionally, I was responsible for the initial product design phase, delegating and coordinating tasks between team members, creating and managing the github repository, creating and supervising the database.

- **What technologies were used ?**

We used JavaFX to develop the desktop application, then we used the same project structure to develop the application with Symfony and transition it into a web application whilst the database was created and maintained in Supabase. The tech stack choices were limited to us by the university professors supervising our project.

- **What's the current status ?**

The academic project is finished and delivered to our university professors but it hasn't been deployed to any platform since we were only asked to provide an application that runs locally on the machine.